

Special right triangles



1 The side opposite the angle measuring 30° .

2 1 unit

3

a $b = \sqrt{3}$

b $\theta = 30^\circ$

4

a

a	1	2	3	4	5
b	1	2	3	4	5
c	$\sqrt{2}$	$2\sqrt{2}$	$3\sqrt{2}$	$4\sqrt{2}$	$5\sqrt{2}$

b x units

c $\sqrt{2}$ units

d The two shortest sides have the same length and the length of the hypotenuse is $\sqrt{2}$ times larger than either of the other side lengths.

5

a

a	1	2	3	4	5
b	$\sqrt{3}$	$2\sqrt{3}$	$3\sqrt{3}$	$4\sqrt{3}$	$5\sqrt{3}$
c	2	4	6	8	10

b $x\sqrt{3}$ units

6

a

i $a = 15$

ii $c = 15\sqrt{2}$

b

i $a = 3$

ii $c = 6$

c

i $a = \frac{4\sqrt{3}}{3}$

ii $c = \frac{8\sqrt{3}}{3}$

7

a

i $b = 19$

ii $c = 19\sqrt{2}$

b

i $b = \frac{3}{4}$

ii $c = \frac{3\sqrt{2}}{4}$

c

i $b = 5\sqrt{2}$

ii $c = 10$

d

i $b = 4\sqrt{3}$

ii $c = 8$

e

i $b = 3\sqrt{3}$

ii $c = 6$

f

i $b = 9\sqrt{6}$

ii $c = 18\sqrt{2}$

8

a

i $a = 3$

ii $b = 3$

b

i $a = 7\sqrt{2}$

ii $b = 7\sqrt{2}$

c

i $a = 2\sqrt{15}$

ii $b = \sqrt{30}$

d

i $a = \frac{9\sqrt{6}}{2}$

ii $b = \frac{9\sqrt{6}}{2}$

e

i $a = 5$

ii $b = 5\sqrt{3}$

f

i $a = \frac{7}{2}$

ii $b = \frac{7\sqrt{3}}{2}$

9 $\frac{13}{2}$

10

a $p = 2\sqrt{3}$ units

b $q = 4\sqrt{3}$ units

c $r = 3$ units

d $s = 3\sqrt{3}$ units

e $(9\sqrt{3} + 3)$ units

11

a $148\sqrt{2}$ pixels

b 148 pixels

12

a $\frac{15\sqrt{2}}{2}$ units

b $30\sqrt{2}$ units

13 $\frac{16\sqrt{3}}{9} \text{ m}^2$

14 36 cm





Additional questions

15 $\sqrt{2}$ cm

16

a 1

b 45°

17

a $\sqrt{3}$

b 60°

18

a 30°

b $(3 + \sqrt{3})$ cm

19

a $x = \frac{\sqrt{30}}{2}$

b $h = 1$

c $x = 2$

d $x = 4\sqrt{6}, y = 4\sqrt{3}$

e $u = 8, v = 4\sqrt{3}$

f $a = \frac{9\sqrt{3}}{2}, b = \frac{9}{2}$

g $x = 7\sqrt{2}, y = 14\sqrt{2}$

h $a = 9\sqrt{2}, b = 6\sqrt{3}, c = 12\sqrt{3}$

20

a 45°

b 60°

21

a 60°

b 30°

c x units

d $x\sqrt{3}$ units

e i False. The hypotenuse of a $30^\circ - 60^\circ - 90^\circ$ triangle is twice the length of the shorter leg.

ii True

iii False. The longer leg of a $30^\circ - 60^\circ - 90^\circ$ triangle is $\sqrt{3}$ times the length of the shorter leg.

iv True



To prove: $c = a\sqrt{2}$

Statements	Reasons
1. $\triangle ABC$ is a $45^\circ - 45^\circ - 90^\circ$ triangle	Given
2. $a^2 + a^2 = c^2$	Pythagorean theorem
3. $2a^2 = c^2$	Simplify
4. $a\sqrt{2} = c$	Square root of both sides
5. $c = a\sqrt{2}$	Symmetric property of equality

24 Yes, Zeki is correct. All isosceles right triangles are similar. An isosceles triangle has two equal interior angle measures, and so if it is also a right triangle the two congruent angles must have a measure of 45° . So any isosceles right triangle has angles $90^\circ, 45^\circ, 45^\circ$. This means that, by the Angle-Angle theorem, all isosceles right triangles are similar.

25 A $30^\circ - 60^\circ - 90^\circ$ triangle can be used to solve the problem. The hypotenuse of a $30^\circ - 60^\circ - 90^\circ$ triangle is twice the length of the shorter leg. The hypotenuse of the bookcase is twice the length of the base so we know the bookcase is a $30^\circ - 60^\circ - 90^\circ$ triangle. The angle formed between the base and the hypotenuse is 60° .